

# Glomerular (Nephrotic) — HIV-Associated Nephropathy (HIVAN)





1. Collapsing FSGS

WHAT YOU SEE

Building sign 'COLLAPSING FSGS' / pathology inset 'COLLAPSING VARIANT FSGS'

WHAT IT ENCODES

HIVAN is the classic COLLAPSING variant of FSGS: light microscopy shows global or segmental collapse of the glomerular capillary tuft with hypertrophy and hyperplasia of overlying visceral epithelial cells (podocytes forming pseudocrescents). It is the most aggressive FSGS variant and is the prototypical lesion of untreated HIV with high-risk APOL1 genotype.

BOARD TRAP

WRONG ANSWER: 'HIVAN shows a membranous (subepithelial spike-and-dome) pattern.' Discriminator: HIVAN is collapsing FSGS, not membranous. Membranous nephropathy CAN occur in HIV/coinfections (HBV, syphilis) as an immune-complex disease, but classic HIVAN is the collapsing-FSGS podocytopathy with tuft collapse and podocyte hyperplasia — pick collapsing FSGS for the rapidly progressive nephrotic HIV patient.

2. Podocytes infected

WHAT YOU SEE

'PODOCYTES INFECTED' near green virus

WHAT IT ENCODES

Direct HIV-1 infection/expresson of viral genes (notably nef and vpr) in podocytes dysregulates the podocyte cytoskeleton, causing dedifferentiation, proliferation, and collapse — the engine of the collapsing injury. This is a podocytopathy driven by viral gene products, not antibody deposition, which is why it tracks with viral replication and APOL1 risk.

BOARD TRAP

WRONG ANSWER: 'immune-complex deposition in the mesangium/subendothelium causes the glomerular damage.' Discriminator: classic HIVAN injury is from DIRECT viral infection of renal epithelial cells (a podocytopathy), with negative or nonspecific immunofluorescence. Immune-complex glomerulonephritis in HIV is a SEPARATE entity (HIV immune-complex kidney disease, 'HIVICK'), not HIVAN.

3. Direct infection of podocytes/tubular epi

WHAT YOU SEE

'DIRECT INFECTION — PODOCYTES + TUBULAR EPI'

WHAT IT ENCODES

HIV directly infects both glomerular podocytes AND tubular epithelial cells; the tubular involvement produces the characteristic microcystic tubular dilation and large echogenic kidneys. So HIVAN is a combined glomerular-and-tubular epithelial disease driven by local viral expression, explaining both the heavy proteinuria and the rapid GFR loss.

BOARD TRAP

WRONG ANSWER: 'damage is purely from circulating immune complexes deposited from the bloodstream.' Discriminator: HIVAN reflects in-situ viral infection of resident kidney epithelium (podocytes + tubular cells), not bystander immune-complex deposition. If the stem emphasizes granular immune deposits, think HIVICK or coinfection-related GN, not HIVAN.

4. Black patients predominantly

WHAT YOU SEE

'BLACK PATIENTS PREDOMINANTLY' under man

WHAT IT ENCODES

HIVAN occurs almost exclusively in patients of African ancestry, because it requires the APOL1 high-risk genotype that is largely confined to that population; it is rare in white or Asian HIV patients. Demographically it strikes Black patients with advanced, poorly controlled HIV (high viral load, low CD4).

BOARD TRAP

WRONG ANSWER: 'HIVAN is equally common across all ethnicities.' Discriminator: HIVAN is strongly enriched in Black patients via APOL1 — a white HIV patient with nephrotic syndrome is far more likely to have an immune-complex GN or another cause than classic HIVAN. Demographics are a deliberate discriminator on boards.

5. APOL1 G1/G2

WHAT YOU SEE

Cup label 'APOL1 G1/G2'

WHAT IT ENCODES

Two APOL1 high-risk alleles (G1 and/or G2 on chromosome 22) confer dramatically increased risk of HIVAN (as well as FSGS and hypertensive nephrosclerosis). HIVAN essentially behaves as a 'second-hit' disease: APOL1 high-risk genotype + uncontrolled HIV viral replication. The same APOL1 alleles drive collapsing FSGS from interferon, lupus, and COVID-19.

BOARD TRAP

WRONG ANSWER: 'HLA-B27 (or another HLA allele) is the predisposing genetic factor.' Discriminator: the HIVAN/collapsing-FSGS susceptibility gene is APOL1 (G1/G2 variants), not HLA-B27 (which is spondyloarthropathies/reactive arthritis). Any 'HLA' answer for HIVAN/FSGS susceptibility is a trap — the answer is APOL1.

6. Heavy proteinuria + AKI

WHAT YOU SEE

Nurse 'HEAVY PROTEINURIA + AKI'

WHAT IT ENCODES

HIVAN classically presents with massive proteinuria PLUS rapidly progressive renal failure — GFR can fall toward ESKD over weeks to a few months if untreated. Importantly, despite heavy proteinuria, blood pressure and edema are often less prominent than expected, and the course is one of rapid decline rather than the indolent course of many other glomerulopathies.

BOARD TRAP

WRONG ANSWER: 'presents with isolated microscopic hematuria and preserved renal function (a nephritic-type picture).' Discriminator: HIVAN is a NEPHROTIC, rapidly-declining disease — heavy proteinuria with rapid GFR loss, not isolated hematuria. Prominent hematuria/RBC casts with HIV should redirect you toward an immune-complex GN (HIVICK, IgA, lupus-like) rather than HIVAN.



7. Often nephrotic

WHAT YOU SEE

'OFTEN NEPHROTIC' caption

WHAT IT ENCODES

Proteinuria in HIVAN is typically in the nephrotic range (>3.5 g/day, often very heavy), reflecting the severe podocyte injury of collapsing FSGS. Nephrotic-range proteinuria in an HIV patient of African ancestry with declining GFR and large kidneys is the classic HIVAN triad.

BOARD TRAP

WRONG ANSWER: 'proteinuria in HIVAN is usually sub-nephrotic / mild.' Discriminator: HIVAN proteinuria is characteristically nephrotic-range and often massive — sub-nephrotic proteinuria argues against classic HIVAN and toward HIVICK, drug effect (e.g., tenofovir tubulopathy), or other CKD causes.

8. Kidneys large on US

WHAT YOU SEE

Banner 'KIDNEYS LARGE on US — even ADVANCED'

WHAT IT ENCODES

On ultrasound, HIVAN kidneys are characteristically ENLARGED and highly echogenic, and they stay large even when renal function is severely impaired — unusual, since most chronic renal disease shrinks the kidneys. The large echogenic kidneys come from tubular microcystic dilation and interstitial expansion.

BOARD TRAP

WRONG ANSWER: 'the kidneys are small and contracted on imaging' (typical CKD picture). Discriminator: HIVAN paradoxically produces normal-to-large echogenic kidneys despite advanced disease — small, scarred kidneys would point to long-standing chronic disease, not HIVAN. Large kidneys + heavy proteinuria + rapid decline in a Black HIV patient = HIVAN.

9. Tubular microcysts

WHAT YOU SEE

'TUBULAR MICROCYSTS' with cystic kidney icon

WHAT IT ENCODES

Microcystic dilation of the renal tubules, filled with proteinaceous casts, is a characteristic histologic feature of HIVAN and a major reason the kidneys are enlarged on imaging. There is also prominent tubulointerstitial inflammation/edema. These tubular changes distinguish HIVAN from a pure glomerular podocytopathy.

BOARD TRAP

WRONG ANSWER: 'no tubular changes occur in HIVAN — it is a purely glomerular disease.' Discriminator: HIVAN prominently involves the TUBULES (microcystic dilation, interstitial inflammation), which is exactly why the kidneys are large/echogenic. The absence of tubular involvement would argue against HIVAN.

10. ART primary treatment

WHAT YOU SEE

Star banner 'PRIMARY TREATMENT' / 'ART antiretroviral therapy'

WHAT IT ENCODES

Combination antiretroviral therapy (cART/ART) is the PRIMARY, first-line treatment for HIVAN — suppressing viral replication halts the podocyte injury and has transformed HIVAN from near-uniform ESKD to a treatable disease; ART should be started promptly regardless of CD4 count. Adjuncts are ACEi/ARB; steroids are only adjunctive/controversial.

BOARD TRAP

WRONG ANSWER: 'high-dose corticosteroids are first-line therapy.' Discriminator: the cornerstone of HIVAN treatment is ART, NOT steroids — controlling the virus treats the cause. Steroids are at most an adjunct in selected refractory cases and are controversial; choosing steroids over ART is the classic HIVAN trap.

11. ACEi/ARB adjunct

WHAT YOU SEE

'ACEi / ARB — ADJUNCT' pill bottle

WHAT IT ENCODES

RAAS blockade with an ACE inhibitor or ARB is a key adjunct in HIVAN to reduce intraglomerular pressure and proteinuria and slow CKD progression, layered on top of ART. Standard antiproteinuric and CKD-protective therapy applies just as in other proteinuric glomerular diseases.

BOARD TRAP

WRONG ANSWER: 'RAAS blockade is contraindicated in HIVAN.' Discriminator: ACEi/ARB are recommended adjuncts in HIVAN (antiproteinuric, renoprotective), not contraindicated — monitor potassium and creatinine. The combination is ART (primary) + RAAS blockade (adjunct).

12. Steroids controversial

WHAT YOU SEE

Doctor with '?' 'STEROIDS — CONTROVERSIAL'

WHAT IT ENCODES

Corticosteroids in HIVAN are adjunctive and CONTROVERSIAL — sometimes considered for severe/rapidly progressive disease or marked interstitial inflammation not responding to ART, but they carry infection risk in an immunocompromised host and are not standard first-line. The established backbone is ART ± RAAS blockade.

BOARD TRAP

WRONG ANSWER: 'corticosteroids are the established standard of care / proven first-line for HIVAN.' Discriminator: steroids are NOT standard first-line — ART is. Steroid use is debated and reserved for selected refractory cases because immunosuppression risks opportunistic infection in HIV. Picking steroids as 'standard of care' is wrong.

13. Virus infecting glomerulus

WHAT YOU SEE

Green HIV virus particle entering building/glomerulus

WHAT IT ENCODES

Active HIV viral replication within renal epithelial cells is central to HIVAN pathogenesis — which is why HIVAN occurs in advanced, uncontrolled HIV (high viral load, low CD4) and why suppressing the virus with ART is therapeutic. The disease essentially requires both ongoing viral expression and the APOL1 high-risk genetic background.

BOARD TRAP

WRONG ANSWER: 'HIVAN is unrelated to active viral replication and can occur with fully suppressed virus.' Discriminator: HIVAN is driven by active viral infection of kidney cells, so it is a disease of uncontrolled HIV and responds to viral suppression. A patient with an undetectable viral load on effective ART developing nephrotic syndrome is more likely to have a different glomerular disease than classic HIVAN.